

PROJECT

5.5

**TESTING VPC
CONNECTIVITY**
NETWORKING SERIES

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Objective

This project tests VPC connectivity by accessing a public EC2 instance from the AWS Console and using it to manage or monitor other resources. It ensures the public server is properly set up for remote access.

It also checks instance to instance communication and confirms that the public subnet has internet access, validating the overall VPC setup, routing, and security configurations.

Services

Amazon EC2 - Runs virtual servers for testing connectivity.

Amazon VPC - Creates a secure network for your resources.

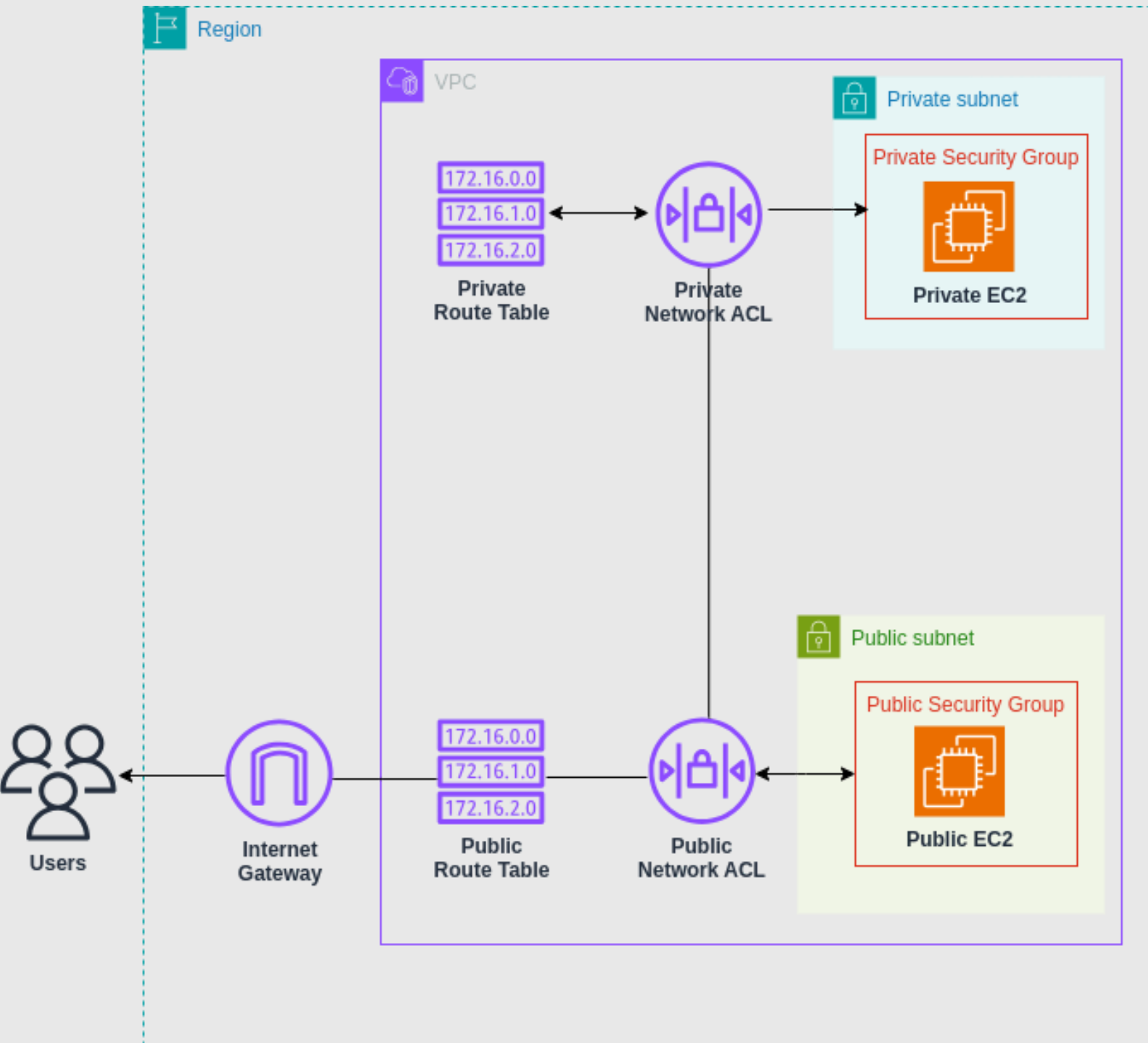
Subnets - Separate public and private zones in VPC

Security Groups - Control traffic between instances.

Internet Gateway - Provides internet access to public subnet.

AWS Console - Lets you manage and connect to instances easily.

Architecture Diagram



The architecture includes a VPC with one public and one private subnet. The public EC2 instance connects to the internet via an internet gateway, while the private EC2 instance communicates only within the VPC. This setup allows secure remote access and internal testing between instances.

Steps to Implement

1. Launch EC2 in Public Subnet:

Launch an EC2 instance in the public subnet with auto assign public IP enabled. This instance will act as your entry point to the VPC. Use an existing key pair to connect securely from your local system.

2. Launch EC2 in Private Subnet:

Create another EC2 instance in the private subnet without assigning a public IP. This instance won't have internet access and will be used to test internal VPC connectivity. Keep both instances in the same VPC for easy communication.

3. Connect to Public Instance:

From the AWS Management Console or via SSH, connect to the public EC2 instance. This helps verify that the public subnet and internet gateway are working. It also sets up a secure path to reach internal instances.

4. Test Internal & Internet Connectivity:

Try pinging or SSHing into the private EC2 using its private IP from the public EC2. Also, test internet access from both instances to confirm the private one is isolated. This ensures subnet, routing, and security settings are correct

Story Time

Public EC2 (The City Gate):

Launch your first EC2 guard at the main gate (public subnet) with an internet badge (public IP). This gatekeeper lets you safely enter your cloud city. Just make sure you have the key (key pair) to talk to them!

Private EC2 (Secret Base):

Now set up a secret agent EC2 deep inside your city walls (private subnet), hidden from the outside world. No internet badge here, just pure stealth! It's perfect for running private tasks away from public eyes.

Connect to the Gatekeeper:

Use your AWS console or terminal to talk to your gatekeeper EC2. If they answer, the city's gate is working fine! This proves your entry route and internet path are secure.

Spy-to-Spy Communication:

Ask the gatekeeper to send a message to the secret agent EC2 using their secret address (private IP). If the message goes through and only the gatekeeper can access the internet, your mission is successful, your city is secure and smartly connected!

BY
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THANK YOU!

